Day 015 | Unit 01 | Teacher plan

### Positive Rational Numbers: Decimals and Division

# Unit 1 Test

Standards: Grade 5 prerequisite, 6.NS.B.3, 6.NS.B.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

## Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

## Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 9, 12

Day 015 | Unit 01 | Unit Test

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## Directions

The first 4 questions (20%) review Grade 5 readiness skills. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

## 01 / Previous unit review | Constructed response

Use long division: 522 ÷ 6. Show the algorithm and check.

## 02 / Previous unit review | Constructed response

Use the standard algorithm: 66.2 + 5.7. Show your regrouping or division steps.

## 03 / Previous unit review | Constructed response

Use the standard algorithm: 68.2 - 7.4. Show your regrouping or division steps.

## 04 / Previous unit review | Constructed response

Calculate 1 × 1/2. Simplify.

## 05 / Constructed response

Use the standard algorithm: 18.7 + 1.5. Show your regrouping or division steps.

## 06 / Constructed response

Use the standard algorithm: 31.6 - 3.4. Show your regrouping or division steps.

## 07 / Constructed response

Use the standard algorithm: 26.6 × 1.5. Show your regrouping or division steps.

## 08 / Constructed response

Use long division: 7,025 ÷ 25. Show the algorithm and check.

## 09 / Constructed response

Use long division: 6,048 ÷ 36. Show the algorithm and check.

## 10 / Constructed response

Use long division: 3,815 ÷ 35. Show the algorithm and check.

## 11 / Constructed response

Use the standard algorithm: 17.92 ÷ 3.2. Show your regrouping or division steps.

## 12 / Multiple choice

Use the standard algorithm: 2.55 ÷ 1.5. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 27/10
B. 17/10
C. 17
D. 7/10

## 13 / Multiple choice

Use the standard algorithm: 7.3 + 2.1. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 94
B. 42/5
C. 47/5
D. 52/5

## 14 / Multiple choice

Use the standard algorithm: 5.58 + 0.68. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 313/5
B. 363/50
C. 313/50
D. 263/50

## 15 / Multiple choice

Use the standard algorithm: 1.968 ÷ 0.24. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 41/5
B. 82
C. 46/5
D. 36/5

## 16 / Model and constructed response

Use long division: 7,129 ÷ 27. Show the algorithm and check.

## 17 / Multiple choice

Use the standard algorithm: 4.3 × 0.73. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 3139/100
B. 2139/1000
C. 3139/1000
D. 4139/1000

## 18 / Multiple choice

Use the standard algorithm: 7.79 + 0.25. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 201/25
B. 226/25
C. 176/25
D. 402/5

## 19 / Multiple choice

Use the standard algorithm: 6.12 + 0.59. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 771/100
B. 571/100
C. 671/100
D. 671/10

## 20 / Multiple choice

Use the standard algorithm: 3.69 - 0.82. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.

A. 287/10
B. 287/100
C. 387/100
D. 187/100

Day 015 | Unit 01 | Answer key

### Positive Rational Numbers: Decimals and Division

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Standards: Grade 5 prerequisite, 6.NS.B.3, 6.NS.B.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

### 1. 87

52 ÷ 6: write 8, subtract 48, remainder 4; bring down the next digit: 42 ÷ 6: write 7, subtract 42, remainder 0. Check: 6 × 87 = 522.

Grade 5 prerequisite | Constructed response

### 2. 71.9

Align decimal points; 66.2 + 5.7 = 71.9.

Grade 5 prerequisite | Constructed response

### 3. 60.8

Align decimal points; 68.2 - 7.4 = 60.8.

Grade 5 prerequisite | Constructed response

### 4. 1/2

1 × 1/2 = 1/2.

Grade 5 prerequisite | Constructed response

### 5. 20.2

Align decimal points; 18.7 + 1.5 = 20.2.

6.NS.B.3 | Constructed response

### 6. 28.2

Align decimal points; 31.6 - 3.4 = 28.2.

6.NS.B.3 | Constructed response

### 7. 39.9

Compute the whole-number product and restore decimal places: 26.6 × 1.5 = 39.9.

6.NS.B.3 | Constructed response

### 8. 281

70 ÷ 25: write 2, subtract 50, remainder 20; bring down the next digit: 202 ÷ 25: write 8, subtract 200, remainder 2; bring down the next digit: 25 ÷ 25: write 1, subtract 25, remainder 0. Check: 25 × 281 = 7025.

6.NS.B.2 | Constructed response

### 9. 168

60 ÷ 36: write 1, subtract 36, remainder 24; bring down the next digit: 244 ÷ 36: write 6, subtract 216, remainder 28; bring down the next digit: 288 ÷ 36: write 8, subtract 288, remainder 0. Check: 36 × 168 = 6048.

6.NS.B.2 | Constructed response

### 10. 109

38 ÷ 35: write 1, subtract 35, remainder 3; bring down the next digit: 31 ÷ 35: write 0, subtract 0, remainder 31; bring down the next digit: 315 ÷ 35: write 9, subtract 315, remainder 0. Check: 35 × 109 = 3815.

6.NS.B.2 | Constructed response

### 11. 5.6

Multiply both numbers by 10: 179.2 ÷ 32 = 5.6. Check: 3.2 × 5.6 = 17.92.

6.NS.B.3 | Constructed response

### 12. B - 17/10

Multiply both numbers by 10: 25.5 ÷ 15 = 1.7. Check: 1.5 × 1.7 = 2.55.

6.NS.B.3 | Multiple choice

### 13. C - 47/5

Align decimal points; 7.3 + 2.1 = 9.4.

6.NS.B.3 | Multiple choice

### 14. C - 313/50

Align decimal points; 5.58 + 0.68 = 6.26.

6.NS.B.3 | Multiple choice

### 15. A - 41/5

Multiply both numbers by 100: 196.8 ÷ 24 = 8.2. Check: 0.24 × 8.2 = 1.968.

6.NS.B.3 | Multiple choice

### 16. 264 R 1

71 ÷ 27: write 2, subtract 54, remainder 17; bring down the next digit: 172 ÷ 27: write 6, subtract 162, remainder 10; bring down the next digit: 109 ÷ 27: write 4, subtract 108, remainder 1. Check: 27 × 264 + 1 = 7129.

6.NS.B.2 | Model and constructed response

## 17. C - 3139/1000

Compute the whole-number product and restore decimal places: 4.3 × 0.73 = 3.139.

6.NS.B.3 | Multiple choice

## 18. A - 201/25

Align decimal points; 7.79 + 0.25 = 8.04.

6.NS.B.3 | Multiple choice

## 19. C - 671/100

Align decimal points; 6.12 + 0.59 = 6.71.

6.NS.B.3 | Multiple choice

## 20. B - 287/100

Align decimal points; 3.69 - 0.82 = 2.87.

6.NS.B.3 | Multiple choice